

Division Decoded

Every graded set, every week — 5 weeks, 25 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Two Kinds of Division

Facts first, then split the dividend into friendly pieces.

✦ Warm-Up 6 ITEMS

$42 \div 6$

$56 \div 8$

$63 \div 9$

$36 \div 4$

$72 \div 8$

$45 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

1. $84 \div 4$ ($80 \div 4 = 20$, then $4 \div 4 = 1$)

.....

2. $96 \div 6$ ($60 \div 6 = 10$, then $36 \div 6 = 6$)

.....

3. $75 \div 5$ ($50 \div 5 = 10$, then $25 \div 5 = 5$)

.....

4. $90 \div 6$ ($60 \div 6 = 10$, then $30 \div 6 = 5$)

.....

5. $78 \div 3$ ($60 \div 3 = 20$, then $18 \div 3 = 6$)

.....

★ Challenge NOT GRADED

C1. Area 96, one side 6. Other side? ($96 \div 6$)

.....

C2. 72 pencils, 8 per box. Boxes? ($72 \div 8$)

.....

The Missing Side

Area and one side are given. Break the area up and hunt the other side.

✦ Warm-Up 6 ITEMS

24 ÷ 3 _____	54 ÷ 6 _____	81 ÷ 9 _____
32 ÷ 8 _____	49 ÷ 7 _____	60 ÷ 5 _____

◆ Core GRADED · SHOW YOUR WORKING

1. 128 ÷ 4 (120 ÷ 4 = 30, then 8 ÷ 4 = 2)

2. 156 ÷ 6 (120 ÷ 6 = 20, then 36 ÷ 6 = 6)

3. 144 ÷ 3 (120 ÷ 3 = 40, then 24 ÷ 3 = 8)

4. 175 ÷ 5 (150 ÷ 5 = 30, then 25 ÷ 5 = 5)

5. 168 ÷ 7 (140 ÷ 7 = 20, then 28 ÷ 7 = 4)

★ Challenge NOT GRADED

C1. Area 252, one side 7. Other side? (210 ÷ 7 = 30, then 42 ÷ 7 = 6)

.....

C2. Bigger answer: 144 ÷ 6 or 168 ÷ 8? Type it. (144 ÷ 6 = 24 and 168 ÷ 8 = 21)

.....

When It Doesn't Fit

Quotient and remainder are two different answers. Read which one is being asked for.

 Warm-Up 6 ITEMS

17 ÷ 5 → remainder

23 ÷ 4 → remainder

38 ÷ 6 → remainder

50 ÷ 7 → remainder

29 ÷ 3 → remainder

45 ÷ 8 → remainder

 Core GRADED • SHOW YOUR WORKING

1. 17 ÷ 5 → quotient (3 fives is 15, and 2 are left)

2. 94 ÷ 4 → quotient (80 ÷ 4 = 20, then 12 ÷ 4 = 3)

3. 94 ÷ 4 → remainder (4 × 23 = 92)

4. 137 ÷ 6 → quotient (120 ÷ 6 = 20, then 12 ÷ 6 = 2)

5. 137 ÷ 6 → remainder (6 × 22 = 132)

 Challenge NOT GRADED

C1. 26 kids, vans hold 6. Vans needed? (Four vans carry 24 – two kids still need a seat)

C2. 26 cookies for 6 friends. Whole cookies each? (This time the leftover stays on the plate)

Chunks Get Tidy

Bigger dividends. Pull out the biggest easy chunk first, then finish.

✦ Warm-Up 6 ITEMS

$120 \div 4$

$150 \div 5$

$240 \div 6$

$350 \div 7$

$180 \div 9$

$160 \div 8$

◆ Core GRADED • SHOW YOUR WORKING

1. $372 \div 3$ ($300 \div 3 = 100$, $60 \div 3 = 20$, $12 \div 3 = 4$)

.....

2. $465 \div 5$ ($450 \div 5 = 90$, then $15 \div 5 = 3$)

.....

3. $728 \div 8$ ($720 \div 8 = 90$, then $8 \div 8 = 1$)

.....

4. $594 \div 6$ ($540 \div 6 = 90$, then $54 \div 6 = 9$)

.....

5. $952 \div 7$ ($700 \div 7 = 100$, $210 \div 7 = 30$, $42 \div 7 = 6$)

.....

★ Challenge NOT GRADED

C1. $845 \div 4 \rightarrow$ quotient ($800 \div 4 = 200$, then $44 \div 4 = 11$)

C2. $845 \div 4 \rightarrow$ remainder ($4 \times 211 = 844$)

.....

Remainder Race & Puzzles

Enrichment day. Every leftover needs a decision — say which one you made.

★ Challenge NOT GRADED

C1. $100 \div 7 \rightarrow$ quotient ($7 \times 14 = 98$)

C2. $100 \div 7 \rightarrow$ remainder ($100 - 98$)

C3. $7 \times \square = 91 \rightarrow \square$ ($70 + 21 = 91$)

C4. $\square \div 6 = 14 \rightarrow \square$ (6×14)

C5. 53 people, 8 per table. Tables? (Six tables seat 48 — round up)

C6. \$53, tickets cost \$8. Tickets? (Drop the leftover — you can't buy part of a ticket)

C7. Smallest number over 50 that 7 divides? (7×8)

C8. $9 \times \square = 108 \rightarrow \square$ ($90 + 18 = 108$)

Week 1 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Two Kinds of Division

OUT OF 11

Warm-Up • 7 | 7 | 7 | 9 | 9 | 9

Core • 21 | 16 | 15 | 15 | 26

Challenge • 16 | 9

1.2 • The Missing Side

OUT OF 11

Warm-Up • 8 | 9 | 9 | 4 | 7 | 12

Core • 32 | 26 | 48 | 35 | 24

Challenge • 36 | 24

1.3 • When It Doesn't Fit

OUT OF 11

Warm-Up • 2 | 3 | 2 | 1 | 2 | 5

Core • 3 | 23 | 2 | 22 | 5

Challenge • 5 | 4

1.4 • Chunks Get Tidy

OUT OF 11

Warm-Up • 30 | 30 | 40 | 50 | 20 | 20

Core • 124 | 93 | 91 | 99 | 136

Challenge • 211 | 1

Fri • Remainder Race & Puzzles

ENRICHMENT

Challenge • 14 | 2 | 13 | 84 | 7 | 6 | 56 | 12

Round It Up

Vans, boats and buses. Two leftovers still force one more of everything.

✦ Warm-Up 6 ITEMS

20 ÷ 5

21 ÷ 5 — the remainder

9 people, vans of 4 — vans needed

12 people, vans of 4

13 people, vans of 4

7 people, boats of 3

◆ Core GRADED • SHOW YOUR WORKING

1. 127 children, buses of 30 — buses needed

.....

2. 98 books, boxes of 12 — boxes needed

.....

3. 53 eggs, cartons of 6 — cartons needed

.....

4. 200 seats, rows of 18 — rows needed

.....

5. 75 guests, tables of 8 — tables needed

.....

★ Challenge NOT GRADED

C1. A bus holds 44. How many for 500 children

.....

C2. If $127 \div 30$ needs 5 buses, how many seats sit empty

.....

Drop It

Whole ones each. The leftover just stays on the plate.

 Warm-Up 6 ITEMS

26 ÷ 6 — whole cookies each

26 ÷ 6 — left on the plate

17 ÷ 5 — whole each

30 ÷ 7 — whole each

30 ÷ 7 — leftover

19 ÷ 4 — whole each

 Core GRADED • SHOW YOUR WORKING

1. \$50 shared by 8 — whole dollars each

2. \$50 shared by 8 — dollars left

3. 145 ÷ 12 — whole each

4. 145 ÷ 12 — leftover

5. 200 ÷ 30 — whole each

 Challenge NOT GRADED

C1. 90 cm of ribbon into 14 cm bows — whole bows

C2. That ribbon — cm wasted

Share It Out

Cut the leftovers up and the remainder becomes a fraction.

Warm-Up 6 ITEMS

$1 \div 2$ as a decimal

$1 \div 4$ as a decimal

$9 \div 2$ — whole part

$9 \div 2$ as a decimal

$7 \div 2$ as a decimal

$3 \div 4$ as a decimal

Core GRADED • SHOW YOUR WORKING

1. $26 \div 4$ as a decimal

2. $17 \div 4$ — type the fraction part as a/b

3. $13 \div 4$ as a decimal

4. \$45 shared by 4 — dollars each

5. $22 \div 5$ as a decimal

Challenge NOT GRADED

C1. 3 pizzas between 4 people — each gets, as a/b

C2. \$100 shared by 8 — dollars each

The Remainder Is the Answer

Clocks and repeating patterns. Sometimes the leftover is the whole point.

✦ Warm-Up 6 ITEMS

$10 \div 3$ — the remainder

$14 \div 7$ — the remainder

$25 \div 4$ — the remainder

Days in a week

$9 \div 7$ — the remainder

$20 \div 6$ — the remainder

◆ Core GRADED • SHOW YOUR WORKING

1. Pattern red, blue, green repeating. The 10th bead — type 1, 2 or 3 for its place in the group

.....

2. $100 \div 7$ — the remainder

.....

3. It is 9 o'clock. 25 hours later, the hour shown

.....

4. $50 \div 12$ — the remainder

.....

5. $365 \div 7$ — the remainder

.....

★ Challenge NOT GRADED

C1. A 4-colour pattern. The 63rd item — its place in the group of 4

.....

C2. $1,000 \div 7$ — the remainder

.....

Snack Run Planning

Real quantities, real people. Every leftover decision written down.

★ Warm-Up 2 ITEMS

24 crackers between 4 — each

20 grapes between 3 — whole each

◆ Core GRADED • SHOW YOUR WORKING

1. A pack of 18 between 5 — whole each

.....

2. That pack — leftover

.....

★ Challenge NOT GRADED

C1. \$12 of snacks split 5 ways — dollars each

.....

C2. You need 7 each for 6 people. Packs of 10 needed

.....

C3. Those packs — snacks left over

.....

Week 2 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Round It Up

OUT OF 11

Warm-Up • 4 | 1 | 3 | 3 | 4 | 3

Core • 5 | 9 | 9 | 12 | 10

Challenge • 12 | 23

2.2 • Drop It

OUT OF 11

Warm-Up • 4 | 2 | 3 | 4 | 2 | 4

Core • 6 | 2 | 12 | 1 | 6

Challenge • 6 | 6

2.3 • Share It Out

OUT OF 11

Warm-Up • 0.5 | 0.25 | 4 | 4.5 | 3.5 | 0.75

Core • 6.5 | $1/4$ | 3.25 | 11.25 | 4.4Challenge • $3/4$ | 12.5

2.4 • The Remainder Is the Answer

OUT OF 11

Warm-Up • 1 | 0 | 1 | 7 | 2 | 2

Core • 1 | 2 | 10 | 2 | 1

Challenge • 3 | 6

Fri • Snack Run Planning

OUT OF 4

Warm-Up • 6 | 6

Core • 3 | 3

Challenge • 2.4 | 5 | 8

Under the Bar

Partial quotients folded into the standard layout, line by line.

✦ Warm-Up 6 ITEMS

$60 \div 3$

$80 \div 4$

$90 \div 3$

$48 \div 4$

$66 \div 6$

$84 \div 4$

◆ Core GRADED • SHOW YOUR WORKING

1. $96 \div 4$

2. $126 \div 3$

3. $175 \div 5$

4. $192 \div 6$

5. $133 \div 7$

★ Challenge NOT GRADED

C1. $288 \div 8$

.....

C2. $441 \div 9$

.....

Three-Digit Dividends

Estimate before you start. Every time, out loud.

✦ Warm-Up 6 ITEMS

$300 \div 3$

$400 \div 4$

$360 \div 6$

$250 \div 5$

$280 \div 7$

$540 \div 9$

◆ Core GRADED • SHOW YOUR WORKING

1. $372 \div 3$

.....

2. $456 \div 4$

.....

3. $595 \div 5$

.....

4. $728 \div 8$

.....

5. $651 \div 7$

.....

★ Challenge NOT GRADED

C1. $936 \div 8$

.....

C2. $864 \div 6$

.....

Zeros in the Quotient

The place nobody remembers to hold. It catches everybody once.

✦ Warm-Up 6 ITEMS

$60 \div 6$

$90 \div 9$

$200 \div 2$

$606 \div 6$

$404 \div 4$

$800 \div 8$

◆ Core GRADED • SHOW YOUR WORKING

1. $618 \div 6$ _____

2. $721 \div 7$ _____

3. $915 \div 3$ _____

4. $832 \div 8$ _____

5. $525 \div 5$ _____

★ Challenge NOT GRADED

C1. $4,080 \div 8$

C2. $6,036 \div 6$

Check by Multiplying

Quotient $\times$ divisor + remainder = dividend. Self-marking from here on.

✦ Warm-Up 6 ITEMS

7×8

$56 \div 8$

$9 \times 6 + 2$

12×4

$48 \div 4$

$11 \times 5 + 3$

◆ Core GRADED • SHOW YOUR WORKING

1. 124×3 — the check for $372 \div 3$

2. $17 \times 6 + 4$

3. $106 \div 6$ — the quotient

4. $106 \div 6$ — the remainder

5. 91×8

★ Challenge NOT GRADED

C1. A quotient of 23 remainder 5 with divisor 9 — the dividend
.....

C2. A quotient of 47 remainder 3 with divisor 12 — the dividend
.....

Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

✦ Warm-Up 2 ITEMS

$42 \div 6$

$90 \div 3$

◆ Core GRADED • SHOW YOUR WORKING

1. $372 \div 3$

2. $618 \div 6$

3. 127 children, buses of 30 — buses needed

4. $26 \div 4$ as a decimal

5. $100 \div 7$ — the remainder

★ Challenge NOT GRADED

C1. A quotient of 23 remainder 5 with divisor 9 — the dividend

.....

Week 3 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Under the Bar

OUT OF 11

Warm-Up • 20 | 20 | 30 | 12 | 11 | 21

Core • 24 | 42 | 35 | 32 | 19

Challenge • 36 | 49

3.2 • Three-Digit Dividends

OUT OF 11

Warm-Up • 100 | 100 | 60 | 50 | 40 | 60

Core • 124 | 114 | 119 | 91 | 93

Challenge • 117 | 144

3.3 • Zeros in the Quotient

OUT OF 11

Warm-Up • 10 | 10 | 100 | 101 | 101 | 100

Core • 103 | 103 | 305 | 104 | 105

Challenge • 510 | 1006

3.4 • Check by Multiplying

OUT OF 11

Warm-Up • 56 | 7 | 56 | 48 | 12 | 58

Core • 372 | 106 | 17 | 4 | 728

Challenge • 212 | 567

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 7 | 30

Core • 124 | 103 | 5 | 6.5 | 2

Challenge • 212

Tests for 2, 5 and 10

The last digit is enough on its own. Say why.

Warm-Up 6 ITEMS

Is 46 divisible by 2 — yes or no

Is 45 divisible by 5

Is 45 divisible by 2

Is 70 divisible by 10

Is 75 divisible by 10

Is 128 divisible by 2

Core GRADED • SHOW YOUR WORKING

1. Is 3,450 divisible by 10 — yes or no

.....

2. Is 1,275 divisible by 5

.....

3. Is 2,346 divisible by 2

.....

4. Numbers below 50 divisible by both 2 and 5

.....

5. Is 999 divisible by 5

.....

Challenge NOT GRADED

C1. A number divisible by 2 and 5 is divisible by

.....

C2. Multiples of 10 below 200 — how many

.....

Tests for 3 and 9

Add the digits — then find out why that ridiculous trick works.

✦ Warm-Up 6 ITEMS

Digit sum of 24

Is 24 divisible by 3 — yes or no

Digit sum of 45

Is 45 divisible by 9

Digit sum of 17

Is 17 divisible by 3

◆ Core GRADED • SHOW YOUR WORKING

1. Digit sum of 738

.....

2. Is 738 divisible by 9 — yes or no

.....

3. Is 1,234 divisible by 3

.....

4. Is 5,346 divisible by 9

.....

5. Digit sum of 9,999

.....

★ Challenge NOT GRADED

C1. Smallest digit d making $45d$ divisible by 9

.....

C2. A number divisible by 9 is always divisible by

.....

Four-Digit Dividends

Same method, longer bar. Do not flinch.

✦ Warm-Up 6 ITEMS

$1,000 \div 2$

$2,000 \div 4$

$3,000 \div 3$

$4,800 \div 8$

$5,600 \div 7$

$1,200 \div 6$

◆ Core GRADED • SHOW YOUR WORKING

1. $3,472 \div 8$

.....

2. $4,536 \div 6$

.....

3. $5,432 \div 4$

.....

4. $6,741 \div 3$

.....

5. $8,127 \div 9$

.....

★ Challenge NOT GRADED

C1. $9,072 \div 8$

.....

C2. $7,595 \div 5$

.....

Two-Digit Divisors

Grade-five ceiling. Estimate, adjust, estimate again.

✦ Warm-Up 6 ITEMS

$40 \div 20$

$60 \div 30$

$90 \div 30$

$80 \div 20$

$120 \div 40$

$150 \div 50$

◆ Core GRADED • SHOW YOUR WORKING

1. $84 \div 12$ _____

2. $96 \div 16$ _____

3. $224 \div 14$ _____

4. $375 \div 25$ _____

5. $391 \div 17$ _____

★ Challenge NOT GRADED

C1. $552 \div 24$

C2. $945 \div 35$

Perfect Numbers

$6 = 1 + 2 + 3$. Its factors add up to itself. Hunt the next one.

★ Warm-Up 2 ITEMS

Factors of 6 below 6 — their sum

Factors of 8 below 8 — their sum

◆ Core GRADED • SHOW YOUR WORKING

1. Factors of 28 below 28 — their sum

.....

2. Factors of 12 below 12 — their sum

.....

★ Challenge NOT GRADED

C1. The second perfect number

.....

C2. The third perfect number

.....

C3. Factors of 496 below 496 — their sum

.....

Week 4 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Tests for 2, 5 and 10

OUT OF 11

Warm-Up • yes | yes | no | yes | no | yes

Core • yes | yes | yes | 4 | no

Challenge • 10 | 19

4.2 • Tests for 3 and 9

OUT OF 11

Warm-Up • 6 | yes | 9 | yes | 8 | no

Core • 18 | yes | no | yes | 36

Challenge • 0 | 3

4.3 • Four-Digit Dividends

OUT OF 11

Warm-Up • 500 | 500 | 1000 | 600 | 800 | 200

Core • 434 | 756 | 1358 | 2247 | 903

Challenge • 1134 | 1519

4.4 • Two-Digit Divisors

OUT OF 11

Warm-Up • 2 | 2 | 3 | 4 | 3 | 3

Core • 7 | 6 | 16 | 15 | 23

Challenge • 23 | 27

Fri • Perfect Numbers

OUT OF 4

Warm-Up • 6 | 7

Core • 28 | 16

Challenge • 28 | 496 | 496

Missing-Digit Division

Reason backwards from the quotient to the digits under the bar.

✦ Warm-Up 6 ITEMS

$?\div 4=6$

$?\div 7=8$

$36\div ?=9$

$81\div ?=9$

$?\div 5=12$

$48\div ?=6$

◆ Core GRADED • SHOW YOUR WORKING

1. $?\div 6=103$

.....

2. $372\div ?=124$

.....

3. $?\div 8=91$

.....

4. $?\div 12=15$

.....

5. $595\div ?=119$

.....

★ Challenge NOT GRADED

C1. A 3-digit number $\div 7 = 43$ exactly — the number

.....

C2. $?\div 9 = 47$ remainder 3

.....

Snack Run

Real quantities, real people, every leftover call written down.

Warm-Up 6 ITEMS

30 crackers between 5 — each

24 grapes between 4 — each

18 between 6 — each

25 between 5 — each

40 between 8 — each

21 between 7 — each

Core GRADED • SHOW YOUR WORKING

1. A 44-pack between 6 — whole each

.....

2. That pack — leftover

.....

3. \$18 split 4 ways — dollars each

.....

4. You need 5 each for 9 people. Packs of 12 needed

.....

5. Those packs — snacks left over

.....

Challenge NOT GRADED

C1. \$25 buys packs at \$4 — whole packs

.....

C2. That purchase — dollars change

.....

Remainder Defence

One division, four right answers. Say which one the question wanted.

Warm-Up 6 ITEMS

27 ÷ 4 — the whole part

27 ÷ 4 — the remainder

27 ÷ 4 as a decimal

27 children in cars of 4 — cars needed

27 cookies between 4 — whole each

27 ÷ 4 rounded down

Core GRADED • SHOW YOUR WORKING

1. 53 ÷ 8 — the whole part

2. 53 people in vans of 8 — vans needed

3. 53 ÷ 8 as a decimal

4. 53 ÷ 8 — the remainder

5. \$53 shared by 8 — whole dollars each

Challenge NOT GRADED

C1. A 100-cm ribbon into 15-cm pieces — whole pieces

C2. That ribbon — cm left

Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

 Warm-Up 6 ITEMS

63 ÷ 9

120 ÷ 4

Is 45 divisible by 9 — yes or no

26 ÷ 6 — the remainder

81 ÷ 9

200 ÷ 5

 Core GRADED • SHOW YOUR WORKING

1. 456 ÷ 4

.....

2. 618 ÷ 6

.....

3. 3,472 ÷ 8

.....

4. 84 ÷ 12

.....

5. 100 ÷ 7 — the remainder

.....

 Challenge NOT GRADED

C1. 552 ÷ 24

.....

C2. ? ÷ 9 = 47 remainder 3

.....

Mission 02 Test

Twelve items plus the Big Question, answered out loud.

✦ Warm-Up 2 ITEMS

$48 \div 6$

$150 \div 5$

◆ Core GRADED • SHOW YOUR WORKING

1. $372 \div 3$

2. $728 \div 8$

3. $915 \div 3$

4. $3,472 \div 8$

5. $96 \div 16$

6. Is 5,346 divisible by 9 — yes or no

7. 127 children, buses of 30 — buses needed

8. $26 \div 4$ as a decimal

★ Challenge NOT GRADED

C1. $945 \div 35$

.....

C2. A quotient of 23 remainder 5 with divisor 9 — the dividend

.....

Week 5 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

5.1 • Missing-Digit Division

OUT OF 11

Warm-Up • 24 | 56 | 4 | 9 | 60 | 8

Core • 618 | 3 | 728 | 180 | 5

Challenge • 301 | 426

5.2 • Snack Run

OUT OF 11

Warm-Up • 6 | 6 | 3 | 5 | 5 | 3

Core • 7 | 2 | 4.5 | 4 | 3

Challenge • 6 | 1

5.3 • Remainder Defence

OUT OF 11

Warm-Up • 6 | 3 | 6.75 | 7 | 6 | 6

Core • 6 | 7 | 6.625 | 5 | 6

Challenge • 6 | 10

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 7 | 30 | yes | 2 | 9 | 40

Core • 114 | 103 | 434 | 7 | 2

Challenge • 23 | 426

Fri • Mission 02 Test

OUT OF 10

Warm-Up • 8 | 30

Core • 124 | 91 | 305 | 434 | 6 | yes | 5 | 6.5

Challenge • 27 | 212